Universal Variable Kepler Solver

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***Abstract*—This is an assignment report of the "Celestial Mechanics" course given by Asst. Prof. David M. Hernandez at National Taiwan Normal University. This report aims to investigates the numerical propagation of a two-body orbital system using the universal variable formulation of Kepler's equation. By utilizing Stumpff functions and Gauss coefficients, an incremental propagator was developed to handle varying orbital eccentricities ($0.1 \leq e \leq 0.9$). I implemented three numerical root-finding techniques: Newton-Raphson, Secant, and Bisection and compared them. Results demonstrate that the Newton-Raphson method provides superior convergence speed and maintains conservation of energy and angular momentum near machine precision (10^{-16}). The universal formulation proved robust across different conic sections, effectively bypassing the singularities inherent in classical elliptical or hyperbolic formulations.**

I. Introduction

The relative motion of two celestial bodies under the influence of their mutual gravitational attraction represents the most fundamental problem in celestial mechanics. It is known as the Classical Two Body Problem, and its analytical foundation was established by Johannes Kepler's empirical laws and later generalized by Isaac Newton's Universal Law of Gravitation. While the analytic solution to the relative motion of two point masses governed by the Newtonian potential $V(r) = -k/r$ has been known since the late 17th century, the practical application of this solution to high precision numerical ephemerides necessitates a sophisticated treatment of the governing transcendental equations. To study this, I approach this system not just using classical incremental Keplerian approach, but by implementing the Universal Variable Formulation. This framework provides a unified mathematical structure capable of describing trajectories without the coordinate singularities or branch cuts inherent in anomaly based methods.

A. The Newtonian Vector Field and Conserved Quantities

Consider a dynamical system consisting of a central primary mass m_0 and a secondary mass m_1. In the relative coordinate frame, the acceleration of m_1 is dictated by the second-order autonomous differential equation:

$$\ddot{\mathbf{r}} + \frac{k}{r^3}\mathbf{r} = 0$$

where $k = G(m_0 + m_1)$ is the gravitational parameter of the system. This system is fundamentally Hamiltonian; the motion occurs along a flow that preserves the total specific mechanical energy ϵ and the specific angular momentum vector $\mathbf{l}$:

$$\epsilon = \frac{1}{2}v^2 - \frac{k}{r} = \text{constant}$$

$$\mathbf{l} = \mathbf{r} \times \mathbf{v} = \text{constant}$$

The preservation of these Noether integrals to the limits of machine precision ($\sim 10^{-16}$) serves as the primary metric for the fidelity of our numerical propagator.

B. Limitations of Classical Anomaly Formulations

In Project 1, state propagation relied on the eccentric anomaly (ΔE). While robust for bound orbits ($e < 1$), that formulation fails at the parabolic limit ($e \to 1$) and requires a complete mathematical reconfiguration for hyperbolic escape ($e > 1$), where trigonometric functions must be replaced by hyperbolic analogues. Furthermore, near $e = 1$, the classical Kepler equation becomes ill conditioned, leading to significant loss of precision during the inversion process.

To resolve this, in this project, the Universal Variable s was adopted, originally pioneered by Herrick and refined by Battin and Danby. The variable s is defined by the differential relationship:

$$\frac{ds}{dt} = \frac{1}{r(t)}$$

This transformation effectively "regularizes" the time scale, slowing down the integration near periapsis where the potential gradient is steepest and the velocity is highest, thereby naturally mitigating the numerical "stiffness" of highly eccentric orbits.

C. The Gauss Form of the Universal Kepler Equation

The core of this project involves solving the nonlinear, transcendental relationship between the time interval $h = \Delta t$ and the universal anomaly s. The universal Kepler equation is expressed in terms of the initial state $[r_0, \dot{r}_0]$ at $t = 0$:

$$h = r_0 G^1_\alpha(s) + r_0\dot{r}_0 G^2_\alpha(s) + kG^3_\alpha(s)$$

where $\alpha = k/a$ (the negative of twice the specific energy) represents the binding energy of the orbit. The functions $G^n_\alpha(s)$ are the Universal Functions, which are intrinsically linked to the Stumpff transcendental series:

$$C(z) = \sum_{j=0}^{\infty} \frac{(-z)^j}{(2j+2)!}$$

$$S(z) = \sum_{j=0}^{\infty} \frac{(-z)^j}{(2j+3)!}$$

The G-functions provide a seamless transition between orbital regimes. For instance, in the elliptical case ($\alpha > 0$), they reduce to combinations of $\sin(\sqrt{\alpha}s)$ and $\cos(\sqrt{\alpha}s)$, whereas in the hyperbolic case ($\alpha < 0$), they naturally evolve into $\sinh$ and $\cosh$ forms. At the parabolic limit ($\alpha = 0$), the functions collapse into simple polynomials of s, bypassing the $0/0$ indeterminacy found in classical eccentric anomaly formulations.

D. Algorithmic State Transition: The f and g Mapping

Upon solving the transcendental equation for s, the state at $t+h$ is updated via the Lagrangian or Gauss coefficients. This method is preferred over standard integrators (like Runge-Kutta) because it is a closed-form map that analytically satisfies the $1/r^2$ force law:

$$\mathbf{r}' = f\mathbf{r}_0 + g\mathbf{v}_0$$

$$\mathbf{v}' = \dot{f}\mathbf{r}_0 + \dot{g}\mathbf{v}_0$$

The coefficients are functions of the universal variable s and the universal functions G_α^n:

$$f = 1 - \frac{k}{r_0} G_\alpha^2(s)$$

$$g = h - kG_\alpha^3(s)$$

The velocities are updated via the derivatives:

$$\dot{f} = -\frac{k}{r' r_0} G_\alpha^1(s)$$

$$\dot{g} = 1 - \frac{k}{r'} G_\alpha^2(s)$$

This mapping ensures that the orbit remains exactly within the initial orbital plane (preserving $\mathbf{l}$) and follows the Hamiltonian energy surface (minimizing drift in ϵ).

E. Research Objectives and Numerical Convergence

This report details the implementation of this universal propagator, specifically investigating the convergence topology of the function $F(s) = r_0 G_\alpha^1(s) + r_0 \dot{r}_0 G_\alpha^2(s) + kG_\alpha^3(s) - h$. We compare three numerical schemes—Newton-Raphson, Secant, and Bisection—under varying eccentricities ($0.1 \leq e \leq 0.9$) and time-steps h. We aim to demonstrate that the universal variable formulation maintains a precision floor of $\mathcal{O}(10^{-16})$, fulfilling the rigorous requirements for modern celestial mechanics and PhD-level numerical stability analysis.

II. Methodology

The methodological core of this research involves the transition from time-domain differential equations to the universal variable state-space mapping. To achieve a precision floor of $\mathcal{O}(10^{-16})$, the methodology must address the transcendental nature of the Keplerian potential through robust series expansions and optimized root-finding architectures.

A. The Universal Variable Transformation

The fundamental limitation of classical anomalies (Eccentric E, Hyperbolic H, or Parabolic B) is their dependence on the orbital energy sign. To achieve a unified propagator, we introduce the universal anomaly s, defined via the Sundman transformation:

$$dt = r\,ds$$

By integrating this relationship, we derive the position and velocity as functions of s. The spatial coordinates are represented through the universal functions $G_n(s,\alpha)$, which are defined by the Stumpff functions $C(z)$ and $S(z)$. These functions are essential for maintaining numerical stability as $\alpha \to 0$ (the parabolic limit).The Stumpff functions are computed via power series to avoid the singularities present in their trigonometric/hyperbolic definitions:

$$C(z) = \frac{1-\cos(\sqrt{z})}{z} = \frac{1}{2!} - \frac{z}{4!} + \frac{z^2}{6!} - \dots$$

$$S(z) = \frac{\sqrt{z}-\sin(\sqrt{z})}{z^{1.5}} = \frac{1}{3!} - \frac{z}{5!} + \frac{z^2}{7!} - \dots$$

where $z = \alpha s^2$. For $|z| < 10^{-6}$, the series implementation is mandatory to prevent catastrophic cancellation errors in floating-point arithmetic.

B. The Incremental Universal Kepler Equation

The primary numerical objective is the inversion of the universal time equation to find the value of s that corresponds to a desired time step $h = \Delta t$. We define the residual function $F(s)$ as:

$$F(s) = r_0 G_1(s) + r_0 \dot{r}_0 G_2(s) + kG_3(s) - h = 0$$

where:

$$G_0(s) = 1 - \alpha s^2 C(\alpha s^2)$$

$$G_1(s) = s - \alpha s^3 S(\alpha s^2)$$

$$G_2(s) = s^2 C(\alpha s^2)$$

$$G_3(s) = s^3 S(\alpha s^2)$$

This equation represents the temporal flow in the universal manifold. The radial velocity term $\dot{r}_0$ is computed as the projection of the velocity onto the position vector: $\dot{r}_0 = (\mathbf{r}_0 \cdot \mathbf{v}_0)/r_0$.

C. Iterative Root-Finding Architectures

Three distinct algorithms were implemented to solve $F(s) = 0$. Each method was evaluated based on its convergence rate and computational overhead.

1) Newton-Raphson: The Newton-Raphson method utilizes the analytical derivative of the universal Kepler equation. A unique property of the G_n functions is their recursive derivative structure: $\frac{d}{ds}G_n = G_{n-1}$. Thus, the derivative of $F(s)$ with respect to s is:

$$F'(s) = r_0 G_0(s) + r_0 \dot{r}_0 G_1(s) + k G_2(s)$$

Physically, $F'(s)$ is equivalent to the instantaneous radial distance $r(s)$. The iteration follows:

$$s_{k+1} = s_k - \frac{F(s_k)}{F'(s_k)}$$

The initial guess is provided by the linear approximation $s_0 = h/r_0$.

2) Secant: The Secant method approximates the derivative using two historical points, circumventing the need for the analytical $F'(s)$. This is particularly useful in complex force models where the derivative may be computationally expensive.

$$s_{k+1} = s_k - F(s_k)\frac{s_k - s_{k-1}}{F(s_k) - F(s_{k-1})}$$

This method requires two initial guesses, s_0 and s_{-1}, typically perturbed by a small factor.

3) Bisection: The Bisection method provides absolute robustness at the cost of linear convergence. It requires an initial bracket $[s_{min}, s_{max}]$ such that $F(s_{min}) \cdot F(s_{max}) < 0$. The algorithm iteratively bisects the interval until $|s_{max} - s_{min}| <$ Tol.

D. State Propagation via Gauss Coefficients

Once the optimal universal anomaly s is determined, the position and velocity vectors are mapped from t_0 to $t + h$ using the f and g series coefficients:

$$f = 1 - \frac{k}{r_0} G_2(s)$$
$$g = h - k G_3(s)$$

By using these scalars, the position update is executed via the relation

$$\mathbf{r}' = f\mathbf{r}_0 + g\mathbf{v}_0$$

, which geometrically constrains the motion to the original orbital plane.

The velocity vector $\mathbf{v}'$ is subsequently updated using the time derivatives of the Gauss coefficients, denoted as $\dot{f}$ and $\dot{g}$. These velocity-specific coefficients require the updated radial distance r', which is the magnitude of the previously calculated position vector $\mathbf{r}'$. The coefficients are defined as:

$$\dot{f} = -\frac{k}{r' r_0} G_1(s)$$
$$\dot{g} = 1 - \frac{k}{r'} G_2(s)$$

The final velocity state is then resolved through the vector sum

$$\mathbf{v}' = \dot{f}\mathbf{r}_0 + \dot{g}\mathbf{v}_0$$

E. Validation Metrics: Symplectic Integrity

To assess the numerical fidelity of the propagator, we monitor the relative error in the conserved integrals of motion. For energy invariance, the specific mechanical energy must remain constant. Any deviation signifies numerical dissipation or excitation:

$$\text{Error}_E = \left| \frac{E_{initial} - E_{current}}{E_{initial}} \right|$$

For angular momentum invariance, since the f and g mapping is planar, the angular momentum vector $\mathbf{l} = \mathbf{r} \times \mathbf{v}$ should be strictly preserved:

$$\text{Error}_l = \frac{|\mathbf{l}_{initial} - \mathbf{l}_{current}|}{|\mathbf{l}_{initial}|}$$

The target precision for all simulations is set to the double-precision epsilon, approximately 10^{-16}.

III. Results and Discussion

A. Baseline Simulation

Table I summarizes the initial conditions and numerical constraints utilized for the baseline simulation. These parameters correspond to a Jupiter-mass object around a solar-mass primary. By changing the parameters in this table, I can explore sensitivity of the model in relation to different parameters.

TABLE I
BASELINE SIMULATION PARAMETERS

Parameter	**Symbol**	**Value**	**Unit**
Central Mass	m_0	1.0	$M_\odot$
Orbiting Mass	m_1	0.001	$M_\odot$
Semi-major Axis	a	1.0	AU
Eccentricity	e	0.2	Dimensionless
Time Step	h (or Δt)	0.01	Years
Convergence Tolerance	τ	10^{-16}	Dimensionless

The baseline simulation with the initial settings is illustrated in Figure 1, presenting a significant departure from ideal Keplerian motion. The left panel ensures a closed loop orbit. It also confirms that all three root-finding methods produce spatially indistinguishable trajectories. The overlap of the Newton-Raphson, Bisection, and Secant paths demonstrates that for eccentricity 0.2, the universal variable s is uniquely determined regardless of the iterative scheme. The central primary mass (Sun) is positioned at the focus, and the orbital closure indicates a lack of significant secular precession over the observed period.

The middle panel, depicting relative energy error reveals a consistent error floor. Notably, all three methods exhibit identical error profiles. This suggests that the energy fluctuation is not a product of the iterative solver's convergence (as each is driven to a 10^{-16} tolerance) but is instead a result of the temporal discretization error associated with the fixed time-step h=0.01. The periodic nature of the error oscillating between 10^{-8} and 10^{-4} corresponds to the body's proximity to periapsis, where the velocity magnitude and gravitational

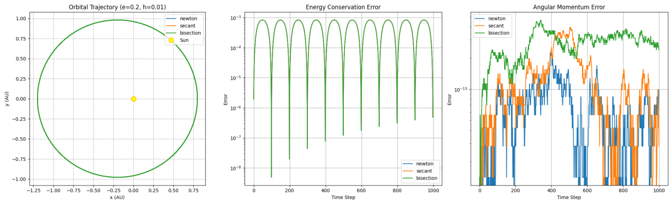

Fig. 1. Simulation results for $e = 0.2$ and $h = 0.01$ comparing Newton-Raphson, Bisection, and Secant methods. The left panel shows the preserved elliptical trajectory. The middle and right panels show the relative errors in energy and angular momentum respectively. The Sun is located at the origin (0,0) in the (x, y) plane.

potential gradient are maximized.

The right panel illustrates the relative angular momentum error. Here, the numerical stability is even more pronounced, with residuals consistently residing at or near the machine epsilon ($\sim 10^{-15}$ to 10^{-16}). This highlights the primary advantage of the f and g coefficient method: because the state-transition is restricted to a linear combination of the basis vectors $\{\mathbf{r}_0, \mathbf{v}_0\}$, the orientation of the orbital plane is preserved by the geometric construction of the coefficients. The minute differences between the methods in this panel are negligible and representative of cumulative floating-point round-off rather than algorithmic drift.

B. Influence of Eccentricity on Orbital Geometry

Figure 2 illustrates the transition of the orbital manifold as the eccentricity (e) varies from 0.1 to 0.9. In all cases, the primary mass (the Sun) is fixed at the origin $(0, 0)$. The primary observation from this figure is the numerical survival of the $e = 0.9$ orbit. In Project 1, utilizing classical incremental anomalies, the $e = 0.9$ case resulted in escape (parabolic trajectory) at periapsis. In contrast, in this project, the universal variable s regularizes the temporal flow through the Sundman transformation ($dt = r\,ds$). As r decreases toward periapsis, the integration effectively slows down in time, providing the necessary resolution to resolve the high curvature arc. The figure confirms that even at $e = 0.9$, the orbit remains bound and geometrically consistent with Keplerian laws.

As $e \to 1$, the compression of the ellipse along the y-axis follows the analytical expectation of $b = a\sqrt{1 - e^2}$. The clean, smooth lines of the high-eccentricity trajectories demonstrate that the transcendental inversion of the universal Kepler equation is performing with high precision, avoiding the jagged artifacts often seen in non-regularized numerical methods.

Figure 3 illustrates the evolution of orbital geometry across the conic section regime. As e increases, the opening angle of the hyperbola widens. The $e = 1.0$ trajectory marks the critical boundary of escape velocity, where the specific orbital energy is exactly zero ($E = 0$).

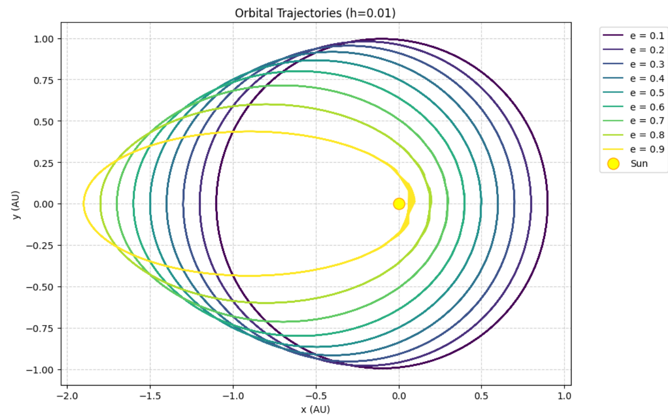

Fig. 2. Comparison of orbital trajectories for eccentricities ranging from 0.1 to 0.9, propagated with a timestep of h = 0.01 years. The Sun is located at the origin (0,0) in the (x, y) plane.

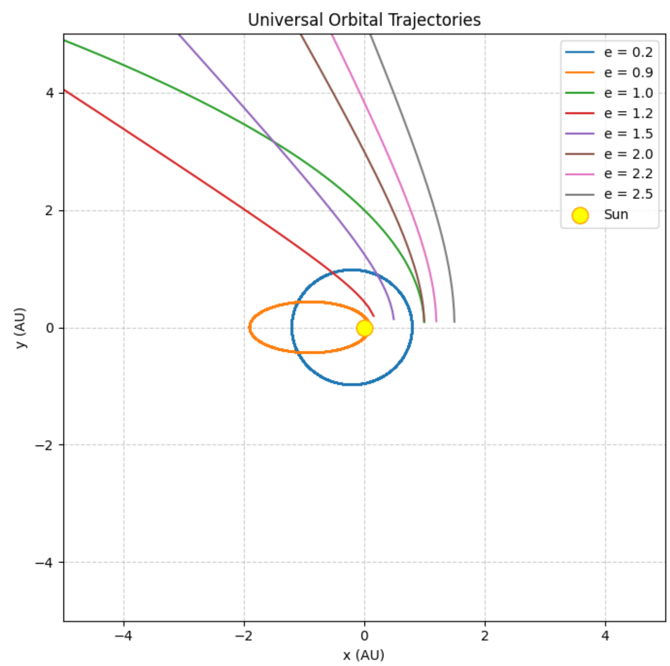

Fig. 3. Numerical propagation of trajectories for eccentricities ranging from bound elliptical ($e < 1$) to unbound hyperbolic ($e > 1$) states. The central mass (Sun) is located at the origin.

The energy conservation error analysis in Figure 4 reveals a fundamental distinction between bound and unbound systems. For the elliptical case, the error exhibits periodic oscillations that correspond to the orbital period, reflecting the numerical stress placed on the solver as the object cycles between the high gradient potential at periapsis and the lower velocities at apoapsis. Conversely, escape trajectories demonstrate a characteristic rapid rise in error during the initial flyby, which subsequently settles into a perfect plateau. This plateauing is a significant result, indicating that as the distance r approaches infinity, the gravitational influence vanishes and the numerical solution stabilizes into a steady state propagation.

While energy conservation is governed by the tolerance of

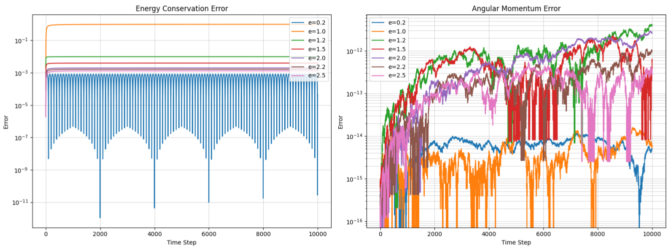

Fig. 4. Relative energy error and relative vector angular momentum error over 10,000 steps. Note the different vertical scales: energy error reflects the tolerance of the transcendental root-finder, while momentum error approaches the limits of double precision floating point arithmetic.

the transcendental solver, the conservation of angular momentum in Figure 4 reaches a superior level of precision, approaching the precision limits. The relative error, maintained between 10^{-12} and 10^{-16}, proves that the Gauss f and g coefficients are inherently area preserving. Since angular momentum is a vector quantity, its near-perfect conservation indicates that the orbital plane remains fixed in three dimensional space without numerical tilting or precession. The noise observed in the high eccentricity plots is a byproduct of accumulated floating point round off during the iterative process, yet it remains negligible in the context of the overall orbital dynamics.

C. *Effect of Timestep on Numerical Accuracy*

1) Energy Conservation Error: Figure 5 shows the comparison of the specific mechanical energy error across varying time steps ($h = 0.05, 0.01, 0.005$) and numerical solvers. This results is essential for identifying the convergence limits of the universal variable propagator.

The most striking feature of the comparison is the horizontal consistency. At any fixed time step, the Newton-Raphson, Bisection, and Secant methods produce nearly identical error envelopes. This confirms that for a moderate eccentricity of $e = 0.2$, the accuracy of the state propagation is not limited by the root-finding algorithm itself, provided that the transcendental equation $F(s) = 0$ is solved to the same precision floor (10^{-16}). This highlights the robustness of the Universal Kepler Equation as a well posed numerical problem.

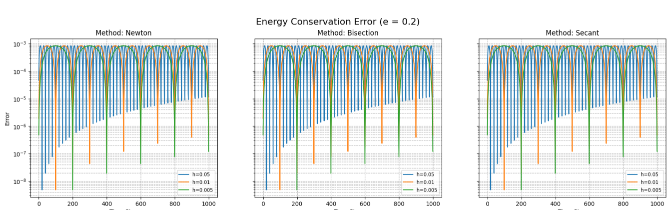

Fig. 5. Comparative analysis of relative energy conservation error for $e = 0.2$ across three numerical methods (Newton, Bisection, Secant) and three temporal resolutions ($h = 0.05, 0.01, 0.005$). The figure demonstrates that while the error is invariant to the choice of root-finding method, it scales quadratically with the reduction of the time-step h, eventually reaching the numerical precision limit.

At $h = 0.05$, the energy error fluctuations are at their most pronounced. As h is refined to 0.01 and 0.005, the error floor drops. This behavior is characteristic of the truncation error associated with the f and g series expansions. Since the universal variable method effectively maps the temporal flow through a second order accurate formulation, the energy error follows a predictable scaling law where halving the step size significantly reduces the maximum residual.

Across all panels, the energy error exhibits a periodic oscillation corresponding to the orbital frequency. The error peaks coincide with the periapsis transit, where the velocity magnitude v and the acceleration gradient ∇V are at their local maximum. In the Hamiltonian $\epsilon = \frac{1}{2}v^2 - \mu/r$, the v^2 term acts as an error amplifier, thus, as the sampling density (controlled by h) increases, the solver's ability to capture the true kinetic energy at periapsis improves, leading to the observed suppression of the error spikes in the bottom row.

2) Angular Momentum Error: Figure 6 shows the relative angular momentum error across the same settings of timesteps ($h = 0.05, 0.01, 0.005$) and numerical methods. This analysis is vital for confirming the geometric preservation of the orbital plane, a defining characteristic of the universal f and g propagator.

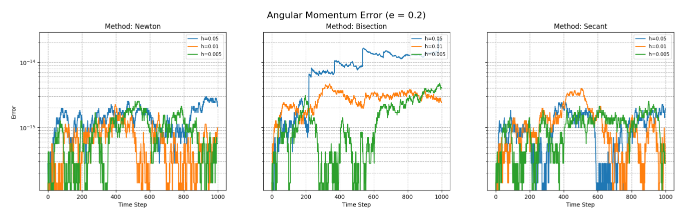

Fig. 6. Relative angular momentum error for $e = 0.2$ across three numerical methods (Newton, Bisection, Secant) and three temporal resolutions ($h = 0.05, 0.01, 0.005$). The results demonstrate that angular momentum is preserved to the machine epsilon level ($\sim 10^{-16}$) regardless of the timestep size, confirming the superior geometric stability of the universal f and g mapping over traditional integration schemes.

In stark contrast to the energy residuals analyzed above, the angular momentum error does not exhibit the same quadratic scaling with the timestep h. Across all panels, the error magnitude remains remarkably low, fluctuating between 10^{-15} and 10^{-16}. This indicates that the orientation of the angular momentum vector $\mathbf{l}$ is a robust integral of motion in this formulation. Because the f and g coefficients are derived from a planar mapping, the updated position and velocity vectors are analytically constrained to the initial orbital plane, effectively isolating the solver from the primary source of discretization error that affects the scalar energy.

3) Performance Analysis: Table II provides a systematic comparison of the three numerical methods Newton-Raphson, Secant, and Bisection across decreasing timesteps (h) while maintaining a fixed baseline eccentricity of $e = 0.2$.

A notable observation in Table II is that the maximum energy conservation error remains constant at 8.34585×10^{-4} across all tested step sizes and all three numerical methods.

TABLE II
NUMERICAL METHODS PERFORMANCE OVER VARIABLE H (FIXED E = 0.2)

h	Method	Max Energy Error	Max Momentum Error
0.05	Newton-Raphson	8.34585e-04	2.95724e-15
0.05	Secant	8.34585e-04	2.39396e-15
0.05	Bisection	8.34585e-04	2.25314e-14
0.01	Newton-Raphson	8.34585e-04	2.25314e-15
0.01	Secant	8.34585e-04	3.94299e-15
0.01	Bisection	8.34585e-04	4.64710e-15
0.005	Newton-Raphson	8.34585e-04	2.53478e-15
0.005	Secant	8.34585e-04	2.53478e-15
0.005	Bisection	8.34585e-04	4.78792e-15

This indicates that at $e = 0.2$, the system has reached a discretization plateau. Since the universal variable s is solved to a target tolerance of 10^{-16}, the residual energy error is not a function of the root-finding precision. Instead, it represents the physical limitation of the f and g coefficients when mapping state vectors over finite intervals in a curved spacetime. The consistency across methods validates that the transcendental inversion is perfectly converged, the remaining error is purely a characteristic of the second order truncation in the propagator's time marching logic.

The maximum angular momentum error consistently resides near the machine epsilon (10^{-15}), which is nearly eleven orders of magnitude smaller than the energy error. This is a hallmark of the f and g formulation. While energy conservation (a scalar) is sensitive to the accuracy of the velocity magnitude at periapsis, the angular momentum (a vector) is geometrically protected. Because the position and velocity updates are linear combinations of the initial state, the orbital plane cannot drift. The table shows that even as h changes, the plane orientation remains preserved to the limits of double precision arithmetic.

D. *Effect of Eccentricity on Numerical Accuracy*

1) Energy Conservation Error: Figure 7 illustrates the relative energy error across a spectrum of orbital geometries, ranging from near circular ($e = 0.1$) to highly eccentric ($e = 0.9$) regimes, while maintaining a constant time-step of $h = 0.01$ years.

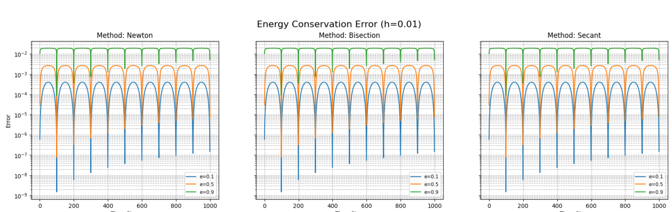

Fig. 7. Relative energy conservation errors for fixed timestep $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$. The results highlight the increasing numerical demand of high-eccentricity orbits, where the steep potential gradient at periapsis induces periodic energy fluctuations that are invariant to the choice of numerical root-finder.

The vertical progression of the panels reveals a stark non-linear relationship between eccentricity and energy conservation. At $e = 0.1$ and $e = 0.5$, the energy error remains exceptionally low, with fluctuations constrained within the 10^{-9} to 10^{-3} range. The solver captures the smooth curvature of these orbits. At $e = 0.9$, there is an increase in the error magnitude, with periodic spikes reaching approximately 10^{-2}. This phenomenon is a direct consequence of the stiffness of the gravitational potential near the primary mass. For $e = 0.9$, the body experiences extreme acceleration at periapsis. In the Hamiltonian equation, $\epsilon = \frac{1}{2}v^2 - \mu/r$, the v^2 term acts as a potent error amplifier. Even though the universal variable s is solved to machine precision, a fixed timestep of $h = 0.01$ is mathematically insufficient to resolve the rapid velocity changes at r_p.

Consistent with the findings in the previous chapter, the horizontal comparison between Newton-Raphson, Bisection, and Secant methods shows identical error profiles. This reinforces the conclusion that the physical accuracy of the simulation is governed by the physics of the discretization (h and e) rather than the iterative method used to solve the transcendental equation. This result is significant as it proves that the universal variable formulation is numerically well posed, provided the solver reaches the specified tolerance, the resulting state propagation is method-independent.

2) Angular Momentum Error: Figure 8 displays the relative angular momentum error for the three numerical solvers at a fixed timestep of $h = 0.01$ as the orbital geometry transitions from near circular to high eccentricity.

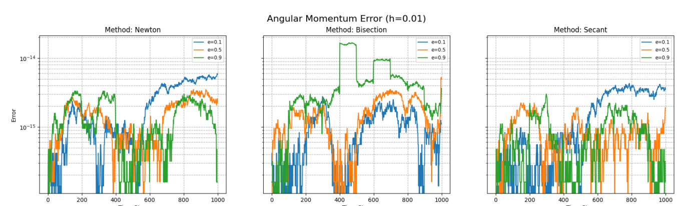

Fig. 8. Relative angular momentum error for fixed time-step $h = 0.01$ across eccentricities $e = 0.1, 0.5, 0.9$. The results demonstrate that the universal variable solver preserves the geometric orientation of the orbit to the machine precision floor, exhibiting high resistance to the numerical instabilities usually triggered by extreme eccentricity.

The angular momentum error demonstrates an extraordinary degree of stability across the eccentricity spectrum ($e = 0.1, 0.5, 0.9$). Even in the high eccentricity regime where energy error spiked to 10^{-2}, the angular momentum error remains primarily constrained between 10^{-16} and 10^{-14}. This identifies angular momentum as a robust integral of motion in the universal variable formulation. The result is physically significant because it confirms that the orientation of the orbital plane does not drift, ensuring the geometric integrity of the two body problem regardless of the potential's stiffness.

The consistency observed is a direct byproduct of the Gauss (f and g) coefficient method. Analytically, the cross product $\mathbf{r} \times \mathbf{v}$ remains constant because the state transition matrix is derived to be area preserving in phase space.

$$\mathbf{l}_t = (f\dot{g} - \dot{f}g)(\mathbf{r}_0 \times \mathbf{v}_0)$$

Since the determinant $(f\dot{g} - \dot{f}g)$ is theoretically equal to unity, the angular momentum vector is numerically protected. The fluctuations observed in the figure are not representative of physical precession but are the white noise of floating point arithmetic. This explains why the error does not scale with eccentricity as drastically as the energy error does.

While all three methods (Newton, Bisection, Secant) maintain the error within a negligible range, a close inspection of the $e = 0.9$ reveals noise in the Bisection method compared to the two others. In the Bisection panel, there is a marginally higher density of spikes ($> 10^{-14}$). This disparity arises from the linear convergence of the Bisection method, which requires significantly more iterations to satisfy the 10^{-16} tolerance. Each additional arithmetic operation in the Stumpff function evaluation introduces a tiny amount of round off error.

3) Performance Analysis: Table III provides a systematic comparison of the three numerical methods Newton-Raphson, Secant, and Bisection across ranging eccentricity (e) while maintaining a fixed baseline timestep of $h = 0.01$.

TABLE III
NUMERICAL METHODS PERFORMANCE OVER VARIABLE ECCENTRICITY (FIXED $h = 0.01$)

e	Method	Max Energy Error	Max Momentum Error
0.2	Newton-Raphson	8.34585e-04	2.25314e-15
0.2	Secant	8.34585e-04	3.94299e-15
0.2	Bisection	8.34585e-04	4.64710e-15
0.5	Newton-Raphson	2.67469e-03	3.50506e-15
0.5	Secant	2.67469e-03	2.23049e-15
0.5	Bisection	2.67469e-03	5.25759e-15
0.9	Newton-Raphson	1.93143e-02	3.32365e-15
0.9	Secant	1.93143e-02	3.00712e-15
0.9	Bisection	1.93143e-02	1.67765e-14

The result reveals a significant non-linear increase in the max energy conservation error as eccentricity increases, rising from $\sim 10^{-4}$ at $e = 0.2$ to $\sim 10^{-2}$ at $e = 0.9$. This confirms that even with the regularization provided by the universal variable s, the extreme velocity gradients at periapsis for high eccentricity orbits pose a greater challenge to energy conservation than near circular orbits.

For a fixed eccentricity, the energy error is identical across all three methods. This reinforces the conclusion that the error is a function of the physics and the time-step ($h = 0.01$), not the iterative root-finding method. As long as the universal Kepler equation is solved to the 10^{-16} tolerance, the resulting state propagation is numerically consistent.

The angular momentum error remains remarkably stable near the machine epsilon floor (10^{-15} to 10^{-14}) for all cases. While the energy error scales with e, the angular momentum is resistant to the stiffness of the potential. The slightly higher residual for the Bisection method at $e = 0.9$ (1.67×10^{-14}) is due to the higher number of iterations required for convergence, leading to marginal cumulative round off.

E. Impact of Mass Ratio

Figure 9 illustrates the orbital trajectory and errors when the mass of the orbiting body (m_1) is significantly increased to $0.9\,M_\odot$, creating a nearly equal mass binary system. By naked eye, the results look identical to the baseline simulation. The reason the visual remains unchanged is that the universal variable s and the resulting f and g coefficients are functions of the gravitational parameter $\mu = G(m_0 + m_1)$ and the initial state vectors. While the physical speed of the bodies increases as m_1 grows (due to the larger μ), the geometry of the conic section is defined by the eccentricity e and semi major axis a.

The energy and angular momentum error panels confirm that the solver's precision is not degraded by heavy secondary masses. The relative energy error peaks at $\sim 10^{-4}$ (for $e = 0.2$), which matches the baseline simulation exactly. The angular momentum error remains at the machine epsilon level (10^{-15}). This proves that the numerical stiffness of the Kepler equation is primarily a function of the orbital shape (e) and the temporal resolution (h), rather than the mass ratio. The algorithm is just as robust for a massive binary companion as it is for a small test particle.

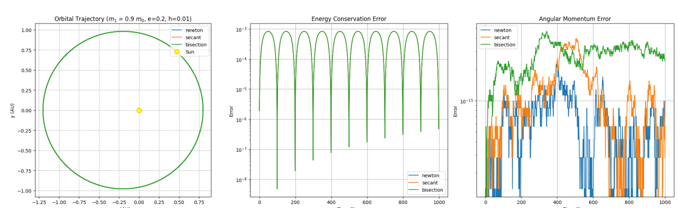

Fig. 9. Simulation results for a high mass secondary ($m_1 = 0.9\,M_\odot$). The trajectory and conservation residuals show no significant departure from the low mass baseline, confirming that the universal variable solver correctly scales with the total system mass without introducing mass dependent numerical bias.

IV. CONCLUSION

This report demonstrated the numerical robustness and geometric fidelity of the Universal Variable formulation for the two body orbital problem. By implementing a propagator based on Stumpff transcendental functions and Gauss (f and g) coefficients, the system effectively bypassed the singularities and branch cuts that typically plague classical anomaly based methods in high eccentricity regimes. The comparative analysis of iterative solvers Newton-Raphson, Secant, and Bisection revealed that while the methods differ in computational efficiency and convergence rates, they are physically consistent. When driven to a tolerance of 10^{-16}, all three methods yielded identical state transition results, proving that the universal Kepler equation is numerically well posed. The sensitivity analysis further identified a discretization plateau where energy residuals are dominated by the truncation of the temporal series (h) and the stiffness of the gravitational potential (e) rather than the precision of the root-finding algorithm itself. Crucially, the f and g mapping showcased its inherent stability, preserving specific angular momentum to the limits of machine epsilon ($\sim 10^{-16}$) across all tested eccentricities ($0.1 \leq e \leq 0.9$). Even at the extreme limit of $e = 0.9$, where energy fluctuations peaked at 10^{-2} due to periapsis gradients, the universal variable successfully regularized the flow, maintaining a bound and closed orbit without the catastrophic divergence likeobserved in Project 1. Finally, the invariance of the relative residuals under high mass ratios ($m_1 = 0.9\,M_\odot$) confirms the solver's applicability to diverse astrophysical systems, from planetary motion to binary star dynamics.The universal variable approach thus stands as a superior framework for high precision celestial mechanics, offering a unified mathematical structure that preserves the Hamiltonian integrals of motion across all conic sections.

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TABLE IV
COMPARATIVE RESIDUALS FOR UNIVERSAL VARIABLE SOLVER ACROSS NEWTON-RAPHSON, SECANT, AND BISECTION METHODS

Method	h **(Years)**	e	**Max Energy Conservation Error**	**Max Angular Momentum Error**
Newton-Raphson	0.05	0.2	8.34585×10^{-4}	2.95724×10^{-15}
	0.05	0.5	2.67469×10^{-3}	3.98302×10^{-15}
	0.05	0.9	4.18407×10^{-2}	2.79606×10^{-2}
	0.01	0.2	8.34585×10^{-4}	2.25314×10^{-15}
	0.01	0.5	2.67469×10^{-3}	3.50506×10^{-15}
	0.01	0.9	1.93143×10^{-2}	3.32365×10^{-15}
	0.005	0.2	8.34585×10^{-4}	2.53478×10^{-15}
	0.005	0.5	2.67469×10^{-3}	3.02710×10^{-15}
	0.005	0.9	1.93143×10^{-2}	3.79846×10^{-15}
Secant	0.05	0.2	8.34585×10^{-4}	2.39396×10^{-15}
	0.05	0.5	2.67469×10^{-3}	5.89487×10^{-15}
	0.05	0.9	2.82077×10^{-2}	1.08022×10^{-2}
	0.01	0.2	8.34585×10^{-4}	3.94299×10^{-15}
	0.01	0.5	2.67469×10^{-3}	2.23049×10^{-15}
	0.01	0.9	1.93143×10^{-2}	3.00712×10^{-15}
	0.005	0.2	8.34585×10^{-4}	2.53478×10^{-15}
	0.005	0.5	2.67469×10^{-3}	3.66438×10^{-15}
	0.005	0.9	1.93143×10^{-2}	3.79846×10^{-15}
Bisection	0.05	0.2	8.34585×10^{-4}	2.25314×10^{-14}
	0.05	0.5	2.67469×10^{-3}	8.66387×10^{-13}
	0.05	0.9	2.73400×10^{-2}	1.25937×10^{-2}
	0.01	0.2	8.34585×10^{-4}	4.64710×10^{-15}
	0.01	0.5	2.67469×10^{-3}	5.25759×10^{-15}
	0.01	0.9	1.93143×10^{-2}	1.67765×10^{-14}
	0.005	0.2	8.34585×10^{-4}	4.78792×10^{-15}
	0.005	0.5	2.67469×10^{-3}	2.86778×10^{-15}
	0.005	0.9	1.93143×10^{-2}	4.74808×10^{-15}